

# Algebra II with Trigonometry

## Chapter 7.2

### Olympiad Problems (GPT-5.4)

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[4582337\\_alg-2trig-h-chp-7-2-note-docx.pdf](#)

### Problems

1. Evaluate exactly:

$$\frac{\sin 80^\circ + \sin 40^\circ + \sin 20^\circ}{\cos 80^\circ + \cos 40^\circ + \cos 20^\circ}.$$

2. Find the exact value of

$$\cos 15^\circ \cos 75^\circ + \sin 15^\circ \sin 75^\circ + \tan 15^\circ \tan 75^\circ.$$

3. Let

$$\alpha = \sin^{-1}\left(\frac{3}{5}\right), \quad \beta = \cos^{-1}\left(\frac{12}{13}\right).$$

Determine the exact value of

$$\tan(\alpha + \beta).$$

4. If

$$\sin \theta + \cos \theta = \frac{1}{3},$$

find the exact value of

$$\sin 3\theta + \cos 3\theta.$$

5. Write

$$5 \sin x - 12 \cos x + 13 \sin\left(x + \frac{\pi}{2}\right)$$

as a single trigonometric function of the form  $R \sin(x + \phi)$ , and hence determine its maximum value.

6. Evaluate exactly:

$$\tan^{-1}(2) + \tan^{-1}(3) - \tan^{-1}(5).$$